

Hoz Serkany

Embedded/C++ Systems Engineer

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LinkedIn | GitHub | Portfolio

SUMMARY

Computer engineering professional building C++ simulation software and embedded Linux systems. Experience spans C++17/CMake/CTest, multithreaded and constrained software, networking and radio interfaces, CI automation, field deployments, testing, and hardware/software integration.

TECHNICAL SKILLS

Core: C++17, C, Python, Java, Bash, object-oriented design, multithreading, debugging, performance optimization

Systems: Linux, embedded Linux, TCP/IP, 5G NR/LTE, 3GPP, Raspberry Pi, microcontrollers, FPGA

Build/Test/Delivery: CMake, CTest, Jenkins, GitHub Actions, Git, Gerrit, Docker, Playwright, Vitest

Domain: CAPE-OPEN (COBIA), process simulation, sensor integration, telemetry, technical documentation

EXPERIENCE

MNPS Inc.

Co-Founder & CTO

Ottawa, Canada

May 2026 – Present

- Lead architecture for a C++17/CMake membrane process simulator covering segment-marching gas separation, multistage cascades with recycle, and plug-flow hollow-fibre models.
- Built multicomponent flowsheets, fibre and module geometry options, and automated CTest regression against literature and patent-aligned stream targets.
- Delivered CAPE-OPEN (COBIA) unit-operation integration through a stable C ABI for industrial hosts such as Aspen Plus and HYSYS.

Sensofusion

Full-Stack Software Developer

Ottawa, Canada

Sep 2025 – Feb 2026

- Supported on-site field deployments, live customer demonstrations, and troubleshooting for command-and-control and sensor-integrated systems.
- Built Node.js services and operator-facing telemetry and geospatial interfaces integrating sensor, system-state, and mission data.
- Reduced streamed payload volume by 70% for duplicated notifications through structural optimization; fixed 30+ production bugs.
- Rebuilt Playwright end-to-end coverage, expanded Vitest/Storybook tests, and collaborated with hardware teams on facility setup and production ramp-up.

Ericsson

Cloud RAN Software Developer (Co-op)

Ottawa, Canada

May 2021 – Apr 2023

- Developed cloud-enabled messaging software for radio equipment running embedded Linux and integrated modules with radio-control systems and legacy interfaces.
- Configured startup and system initialization, added multi-wavelength and older-protocol support, and optimized execution to one processor core by removing extra threads.
- Migrated legacy unit tests from C to C++, automated test configurations, and created Jenkins pipelines and post-merge scripts for module testing.
- Documented system behavior, message flows, and integration points for engineering teams.

Languages of Life

Full-Stack Engineer (Contract)

Remote

Jun 2026 – Present

- Established GitHub Actions quality gates spanning Flutter tests, Deno Edge checks, Postgres RLS tests, and two-session concurrency tests.
- Implemented server-enforced authorization with Postgres grants, RLS, lifecycle RPCs, and optimistic concurrency.

SELECTED PROJECTS

Raspberry Pi Home Assistant Hub: Deploy and administer Raspberry Pi OS, networking, sensor/device integrations, and automations. **16-bit Microprocessor:** Simulated a processor with branching and UART integration using VHDL and assembly.

EDUCATION

Bachelor of Applied Science, Computer Engineering (Innovation Stream), Queen's University

2023